# Chen Cecilia Liu

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***Scholar***: *eNFB3fgAAAAJ*

## Research Interests

Multilinguality, Generalization, Multicultural NLP, Pragmatics-related tasks, Multimodality

## Education

**Technische Universität Darmstadt** (TU Darmstadt) — Darmstadt, Germany
Computer Science — 2021 - Present
European Laboratory for Learning and Intelligent Systems (ELLIS) PhD in NLP (~5% acceptance)
Supervisors: Prof. Iryna Gurevych (TU Darmstadt) / Prof. Anna Korhonen (University of Cambridge)

**University of Cambridge** — Cambridge, United Kingdom
Visiting Research Student @ Language Technology Lab — 2024
Supervisor: Prof. Anna Korhonen

**University of Toronto** — Toronto, Canada
Master of Applied Science, Electrical Engineering — 2011 - 2013
Supervisor: Prof. Brendan Frey

Bachelor of Applied Science (with Honours), Mechanical Engineering — 2006 - 2010

## Relevant Industry Experience

**ElementAI (ServiceNow)** — Toronto, Canada
*Staff Applied NLP Scientist* — Dec 2020 - June 2021

**Wattpad, (acquired by NAVER WEBTOON)** — Toronto, Canada
*Staff Product & Research Data Scientist* — Oct 2020 - Dec 2020
*Senior Product & Research Data Scientist, Tech Lead* — May 2017 - Oct 2020

**Spotify** — Boston, USA
*Machine Learning Engineer* — Sep 2016 - May 2017

**Meta, (acquired by Chan Zuckerberg Initiative)** — Toronto, Canada
*Machine Learning Engineer* — Mar 2014 - Aug 2016

## Honours, Scholarships & Grants

ELIZA Scholarship, DAAD (€ 8700) — 2024
Travel Grant, European Laboratory for Learning and Intelligent Systems (€ 3000) — 2022
Undergraduate Student Research Awards (USRA), NSERC (2x, total $8000) — 2008, 2009
Dean's List, University of Toronto — 2007 - 2010
General Motor of Women in Electrical and Mechanical Engineering Award, University of Toronto (3x, total over

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| $10k) | 2007 - 2009 |
| Mechanical and Industrial Engineering Admission Scholarship, University of Toronto | 2006 |
| Faculty of Applied Sciences and Engineering Admission Scholarship, University of Toronto | 2006 |

## Services

Reviewer/Program committee:

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|---|---|
| NLP | EMNLP (2020, 2022), ACL (2020, 2023), EACL (2021), NAACL (2021), ACL Rolling Review (2022-) |
| ML/AI | NeurIPs (2023, 2024), ICLR (2024, 2025), ICML (2024), COLM (2024, 2025), AAAI (2024) |
| Workshops | RepL4NLP (2024, 2025), WiNLP@EMNLP (2024) |

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| Women in Data Conference, Invited Industry Mentor, Toronto | 2019 |
| Google Go North Conference, Invited Industry Mentor, Toronto | 2017 |
| ECE Department Annual Graduate Students Conference, Organizer, University of Toronto | 2012 |
| Graduate Student Union Student Representative, University of Toronto | 2012 - 2013 |

## Talks

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| Seminar, Invited Talk (Cultural NLP), Ontario Tech University and University of Toronto | 2024 |
| Seminar, Invited Talk (PEFT & Cross-lingual Transfer), Berliner Hochschule für Technik | 2024 |
| CHIA Annual Conference on AI for Good, Poster Presentation, University of Cambridge | 2024 |
| Women in Data Science Conference, Invited Talk, Toronto | 2019 |
| Vector Institute Endless Summer School, Invited Talk, University of Toronto | 2018 |

## Teaching Experience

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| Current Topics in Natural Language Processing, Guest Lecture (Cultural NLP), MBZUAI | 2025 |
| Literature Management for PhD Students, Seminar, TU Darmstadt | 2024 |
| Electrical Fundamentals (ECE101), Tutorial, University of Toronto | 2011 |
| Fundamentals of Computer Programming (APS106), Lab, University of Toronto | 2010 |
| Grade 12 Physics / Chemistry, William International College, Toronto | 2007 |

## Mentorship

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|---|---|
| Nils Kovačić (with Dr. Hiba Arnaout), TU Darmstadt | BSc Thesis, Culturally-aware NLP, Emotion |
| Kai Li, TU Darmstadt | BSc Thesis, Culturally-aware NLP, Multimodality |
| Jonas Stadtmüller, TU Darmstadt | MSc Thesis, Continual Learning, Multimodality |
| Laurenz Kammeyer, TU Darmstadt | BSc Thesis, Prompting |
| Amir Ganiev, University of Toronto | BASc PEY |
| Amanda Steinhauer, University of Toronto | BASc PEY |

## Selected Publications

11. **Chen Liu**, Anna Korhonen and Iryna Gurevych. Cultural Learning-Based Culture Adaptation of Language Models. *Accepted at ACL (to apppear)*, 2025.
10. **Chen Liu**, Iryna Gurevych and Anna Korhonen. Culturally Aware and Culturally Adapted NLP: A Taxonomy and Survey. *Accepted at TACL (in press)*, 2025.

9. **Chen Liu** and Iryna Gurevych. Early Period of Training Impacts Out-of-Distribution Generalization. *Preprint - under review*, 2024. (A workshop version is accepted at HiLD@ICML 2024)
8. **Chen Liu**, Fajri Koto, Timothy Baldwin, and Iryna Gurevych. Are Multilingual LLMs Culturally-Diverse Reasoners? An Investigation into Multicultural Proverbs and Sayings. NAACL-HLT, 2024.
7. **Chen Liu**, Jonas Pfeiffer, Ivan Vulić, and Iryna Gurevych. Improving Generalization of Adapter-Based Cross-lingual Transfer with Scheduled Unfreezing. NAACL-HLT, 2024.
6. **Chen Liu**, Jonas Pfeiffer, Anna Korhonen, Ivan Vulić, and Iryna Gurevych. Delving Deeper into Cross-lingual Visual Question Answering. EACL - *Findings*, 2023.
5. Gregor Geigle, **Chen Liu**, Jonas Pfeiffer, and Iryna Gurevych. One does not fit all! On the Complementarity of Vision Encoders for Vision and Language Tasks. RepL4NLP@ACL, 2023.
4. **Chen Liu***, Gregor Geigle*, Robin Krebs, and Iryna Gurevych. FigMemes: A Dataset for Figurative Language Identification in Politically-Opinionated Memes. EMNLP, 2022.
3. Amir Ganiev, Colton Chapin, Anderson De Andrade, and **Chen Liu**. An Architecture for Accelerated Large-Scale Inference of Transformer-Based Language Models. NAACL - *Industry Track*, 2021.
2. **Chen Liu**, Anderson De Andrade, and Muhammad Osama. 2019. Exploring Multilingual Syntactic Sentence Representations. W-NUT@EMNLP, 2019.
1. **Chen Liu**, Muhammad Osama, and Anderson De Andrade. DENS: A Dataset for Multi-class Emotion Analysis. EMNLP, 2019.